MIL-S-27419(USAF)
24 January 1968

MILITARY SPECIFICATION

STEEL BILLETS, CORROSION RESISTANT, ANNEALED, PREMIUM QUALITY

1. SCOPE

1.1 Scope. This specification covers semifinished forged and annealed corrosion resistant steel billets as preforms of verified soundness quality.

1.2 Classification. Steel billets covered by this specification shall be of the following types and compositions.

Type

Type I - Electric furnace melt process
Type II - Consummable electrode vacuum melt process

Compositions

304L Extra low carbon
321 Titanium stabilized
347 Columbium or columbium plus tantalum stabilized

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

Federal

FED-STD-102	Preservation, Packaging and Packing Levels
FED-STD-151	Metals, Test Methods

Military

MIL-STD-163	Steel Mill Products, Preparation for Shipment and Storage

SPECIFICATION

Military

MIL-I-8950	Inspection, Ultrasonic, Wrought Metals, Process for

FSC 9640

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition shall conform to table I when tested in accordance with applicable test methods.

Table I. Chemical composition limits

Element	Composition limits (percent)			Check Analysis tolerance (percent)
	304L	347	321	
Carbon	0.03 (max)	0.08 (max)	0.08 (max)	+0.01
Manganese	2.00 (max)	2.00 (max)	2.00 (max)	+0.04
Phosphorous	0.025 (max)	0.025 (max)	0.025 (max)	+0.005
Sulfur	0.025 (max)	0.025 (max)	0.025 (max)	+0.005
Chromium	18.0-20.0	17.0 - 20.0	17.0 - 20.0	±0.20
Nickel	8.0 - 11.0	9.0 - 13.0	9.0 - 13.0	±0.15
Silicon	1.00 (max)	1.00 (max)	1.00 (max)	+0.05
Copper	0.50 (max)	0.70 (max)	0.70 (max)	+0.05
Columbium 1/		1.10 (max) 2/	--	+0.05
Titanium		--	0.75 (max) 3/	+0.05
Molybdenum	0.50 (max)	--	--	
Any element not specified	0.20 (max)	0.20 (max)	0.20 (max)	
Sum of elements not specified	0.50 (max)	0.50 (max)	0.50 (max)	
Iron	Remainder	Remainder	Remainder	

1/ Columbium or columbium plus tantalum. The determination of tantalum is not required. The ratio of tantalum to the total of tantalum plus columbium shall not exceed 0.4.

2/ The sum of the columbium content plus the tantalum content shall be not less than 10 times the carbon content.

3/ The ratio of titanium to carbon shall be not less than 5 to 1.

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3.2 Processing.

3.2.1 Production methods. The steel shall be manufactured by the electric furnace melt or the consummable electrode vacuum melt processes as specified (1.2). If the process is not specified, either may be used. Sufficient discard shall be taken from each ingot to ensure freedom from injurious piping and undue segregation.

3.2.2 Forging. The original cross-sectional area of the ingot shall be at least three times the area of the finished forging (a minimum reduction ratio of 3/1), and the produced grain structure of the finished forging shall not interfere with nondestructive testing.

3.2.3 Removal of defects. Surface defects may be removed by machining or grinding provided such removal does not reduce the material below the minimum specified dimensions. Spot ground areas shall be faired smoothly into the surrounding material. When abrasives are used, the process shall not deposit iron on the surface. The removal of a defect shall be verified by the method originally used to detect the defect. Weld repairs are not authorized.

3.2.4 Heat treatment. The material shall be solution treated (annealed) at a minimum temperature of 1900°F, and rapidly cooled. The treatment shall be applied as the final heat treatment on the material as offered for acceptance.

3.2.4.1 Hardness. Each billet shall be checked for hardness after anneal and shall be a reading of not more than BHN 187.

3.2.5 Descaling. All material shall be machined, ground, pickled, or otherwise finished, so that the surface is free from scale and other foreign matter and suitable for testing by ultrasonic inspection. If shot, grit, or vapor blasting is used, it shall be followed by machining or grinding. If abrasives are used, the process shall not deposit iron on the surface.

3.3 Finish. The finish as specified in 3.2.5 shall be suitable for obtaining reproducible results when examined by ultrasonic inspection.

3.4 Shape and dimensions. The shape of the billet as round, ring, or other described shape; and the dimensions as minimum outside diameter, maximum inside diameter, minimum thickness, or other dimensions as applicable shall be as specified (6.2).

3.5 Limits. The following limits shall apply to all measurements of the finished billet at any cross-section at any location.

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3.5.1 Diameters. No outside diameter shall be less than the specified minimum outside diameter, and no inside diameter shall be greater than the specified maximum inside diameter. The measured diameters shall be concentric about a common axis.

3.5.2 Thickness. No thickness shall be less than the specified minimum thickness.

3.5.3 Flatness. The faces shall not be bowed or dished in excess of 1/8 inch.

3.5.4 Corners. The radii at the corners shall not exceed 5/32 inch per inch of thickness.

3.6 Soundness. Each billet shall be nondestructively tested by ultrasonic inspection (4.4.3).

3.7 Defects.

3.7.1 Surface defects. All material shall be uniform in quality, clean and free from seams, cracks, laps, blisters, damaged ends or edges, and other injurious defects as determined by visual or any appropriate nondestructive test method.

3.7.2 Internal defects. All material shall be uniform in quality, sound and free from pipe, laminations, cracks, porosity, and other injurious defects as determined by ultrasonic inspection.

3.8 Identification. One face of each billet shall be legibly die-stamped with the following data:

(a) Specification number and composition.

(b) Producer's name or trademark.

(c) Heat and lot number code.

(d) Composition designation and condition (-O for annealed).

(e) Ultrasonic inspection acceptance stamp and quality level.

3.8.1 In addition, the marked face shall also be stenciled "Premium Quality" with letters 1/2 inch high minimum.

3.9 Certifications. One copy of a certification for the chemical analysis and one copy of a certification for ultrasonic inspection shall be furnished the contracting officer.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection is deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. For the purpose of chemical analysis only, a lot shall consist of all material made from the same heat.

4.3 Sampling. One sample shall be taken from each lot for chemical analysis. The sample may be taken from the ladle or forged products.

4.4 Quality conformance tests.

4.4.1 Chemical analysis.

4.4.1.1 Procedure. The chemical analysis shall meet the composition requirements of table I and shall be performed in accordance with FED-STD-151.

4.4.1.2 Acceptance criteria. When any sample fails to conform to the requirements of table I, the lot represented by that sample shall be rejected. At the option of the supplier, each billet in the rejected lot may be analyzed at the supplier's expense, and each billet that fails to conform to the requirements of table I shall be rejected.

4.4.2 Physical examination.

4.4.2.1 Procedure. All material shall be physically inspected for dimensions and hardness. Inspection aids, tools, and nondestructive tests as necessary shall be used.

4.4.2.2 Acceptance criteria. Any billet that fails to conform to the hardness requirement (3.2.4.1), finish requirements (3.3), dimensional limits (3.5), or surface defects requirements (3.7.1) shall be rejected.

4.4.3 Ultrasonic inspection.

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4.4.3.1 Procedure. All material shall be ultrasonically inspected through 100 percent of its volume by immersion or contact methods in accordance with MIL-I-8950.

4.4.3.2 Acceptance criteria. Any billet that fails to conform to the internal defects (3.7.2) or the ultrasonic quality levels (MIL-I-8950) as follows shall be rejected.

(a) Billets not greater than 2 inches in the thickness dimension shall meet Class A quality level requirements.

(b) Billets greater than 2 inches in the thickness dimension but not greater than 6 inches in the thickness dimension shall meet Class B quality level requirements.

(c) Billets greater than 6 inches in the thickness dimension shall meet Class C quality level requirements.

5. PREPARATION FOR DELIVERY

5.1 Preparation for packaging and packing. Preparation for packaging and packing shall be as follows:

(a) Material shall be clean and free of dirt, chips, or any foreign matter.

(b) Contact preservative shall not be used.

(c) Material shall be segregated as to heat number, type, and size.

5.2 Packaging and packing. The level of packaging and packing in accordance with FED-STD-102 shall be as specified (6.2).

5.2.1 Level A. When Level A (military longterm storage) is specified, material shall be prepared for shipment in accordance with MIL-STD-163 instructions applicable to corrosion resistant steel.

5.2.2 Level C. When Level C (Vendor to user) is specified, material shall be prepared for shipment in accordance with commercial practice to insure carrier acceptance and safe delivery at destination and shall meet, as a minimum, the requirements of carrier rules and regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-163.

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6. NOTES

6.1 Intended use. The material covered by this specification is intended for the manufacture of critical rocket propulsion compounds which are machined to final configuration from the preforms and where minute internal cracking encountered after extensive machining time is involved cannot be tolerated.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and documents should specify the following:

(a) Title, number and date of this specification.

(b) Type and composition desired (1.2).

(c) Shape and dimensions of billets (3.4).

(d) Level of shipment required (5.2).

6.3 When forged billets are ordered, the black-forged dimensions should be 3/8 inch to 1 inch larger than the finished machined dimensions, on surfaces requiring finish machining.

6.4 Composition designations. The use of 304L, 321, and 347 in this specification designates alloys having chemical compositions similar to but not necessarily identical with AISI compositions 304L, 321, and 347 respectively.

Custodians:

Air Force - 12

Review activities:

Air Force - 11, 19

Preparing activity:

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